OFFICIAL TECHNICAL SPECIFICATION

INFINITY LEARNING

BPSC TRE 4.0 Portal & CBT Engine

Complete Technical, Functional, Architectural & Deployment Documentation

TARGET EXAMINATION

BPSC TRE 4.0 (Advt. No. 15/2026) — Bihar School Teacher Recruitment

SYSTEM ARCHITECTURE

Zero-Dependency PHP 8.2+ with PDO SQLite Database Engine

ACCESS CONTROL (RBAC)

3-Tier Matrix: Candidate (Student), Faculty (Teacher), Admin (AIVERSE Master)

RELEASE EDITION

Version 4.0.0 Production Release (November 2026)

Bihar Public Service Commission (BPSC) School Teacher Recruitment Exam Portal System Specification

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1. Executive Summary

Infinity Learning is a web portal designed specifically for candidates, faculty mentors, and administrators preparing for the **Bihar Public Service Commission (BPSC) School Teacher Recruitment Examination (TRE 4.0 / Advt. No. 15/2026)**.

The platform provides:

- **For Candidates (Aspirants)**: Official exam-pattern syllabus breakdown across all 4 school cadres (Primary, Middle, Secondary, Higher Secondary), proctored timed Computer Based Tests (CBT) with authentic negative marking (-0.33), direct downloads of SCERT/NCERT reference books and official Previous Year Question Papers (PYQs), topic revision notes, and real-time District/State merit rankings across Bihar's 38 districts.
- **For Faculty (Teachers)**: Complete authoring tools to upload topic study notes, attach downloadable PDF handouts, upload PYQs, bulk import question banks from Word (`.docx`) files, and publish mock tests with full question bank curation.
- **For Administrators (AIVERSE Master)**: Comprehensive system oversight, teacher onboarding application verification and approvals, fee payment verification ledger, and live question bank governance.

2. Technology Stack & Architectural Principles

The application is engineered with a **pure zero-external-dependency** architecture. It requires only standard PHP 8.2+ and SQLite3 with PDO.

> 📊 **ARCHITECTURE & SCHEMA FLOW SPECIFICATION**

```
graph TD
    Client["Browser Client (Mobile / Tablet / Desktop)"]
    subgraph Web Server ["PHP 8.2 Built-in / Apache / Nginx"]
        Router["Entry Point (index.php)"]
        AuthLayer["Session & Auth Guard (src/Auth.php)"]
        Handlers["API Handlers (*-handler.php)"]
        Views["Modular View Components (views/*)"]
    end
    subgraph Persistence ["Database Layer (PDO SQLite)"]
        DBEngine["src/Database.php"]
        SQLiteDB[("database/bpsc_portal.sqlite")]
    end

    Client --> Router
    Router --> AuthLayer
    AuthLayer --> Views
    Client -->|AJAX / Fetch| Handlers
    Handlers --> DBEngine
    Views --> DBEngine
    DBEngine --> SQLiteDB
```

Core Components:

- **Backend Language**: PHP 8.2+ (strictly typed, zero composer packages required).
- **Database Engine**: SQLite 3 with PDO, WAL mode enabled for rapid concurrent reads.
- **Styling**: Tailwind CSS utility framework loaded via CDN with custom color tokens (`brand-cream`, `brand-orange`, `brand-peachBorder`, `brand-ivory`, `brand-teal`).
- **Typography**: Clean, humanized geometric fonts: *Plus Jakarta Sans* and *Inter* via Google Fonts.
- **Document Parsing**: `Mammoth.js` client-side library for parsing `.docx` Word question papers directly in the browser.
- **Session Management**: Native encrypted PHP sessions (`PHPSESSID`) with secure cookie parameters and role persistence.

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3. Project Directory & File Structure

```
bpsc-tre-portal/
├── .htaccess                    # Apache mod_rewrite & security headers
├── admin-teacher-handler.php    # Endpoint: Faculty approvals & directory management
├── admin.php                    # Admin legacy redirect route
├── index.php                    # Primary portal routing & view assembly
├── leaderboard-handler.php      # Endpoint: Live district/state merit standings
├── login-handler.php            # Endpoint: Multi-role authentication
├── logout.php                   # Endpoint: Session destruction & logout redirect
├── notes-handler.php            # Endpoint: Topic notes & PDF upload management
├── register-handler.php         # Endpoint: Candidate & Teacher onboarding
├── tests-handler.php            # Endpoint: CBT creation, status toggles & submission
├── admin/
│   └── index.php                # Direct admin panel entry script
├── config/
│   ├── config.php               # AppConfig (exam dates, app branding, site URLs)
│   ├── constants.php            # System constants & error codes
│   └── database.php             # SQLite file path & PDO connection settings
├── database/
│   ├── bpsc_portal.sqlite       # Active SQLite database file
│   └── schema.sql               # Complete DDL schema & initial seed dataset
├── src/
│   ├── Auth.php                 # Authentication session manager & role guards
│   ├── Database.php             # PDO SQLite database wrapper with prepared statements
│   ├── LeaderboardManager.php   # Ranking calculation across 38 Bihar districts
```

```
│   └── UserManager.php           # User lifecycle, cadre mapping & teacher onboarding
└── views/
    ├── admin/
    │   └── dashboard.php         # Admin view: verification, ledger, master tables
    ├── auth/
    │   └── gate.php              # Multi-role login & registration modal screens
    ├── cbt/
    │   └── engine.php            # Proctored CBT test interface, timer & scorecard
    ├── layouts/
    │   ├── footer.php            # Core JavaScript application engine & state machine
    │   ├── header.php            # HTML head, responsive viewport, typography & toast
    │   └── nav.php               # Responsive navbar, mobile menu & bottom dock
    ├── modals/
    │   └── modals.php            # All application popups (upload, book, PYQ, payments)
    ├── notes/
    │   └── topic_notes.php       # Full topic study notes view & PDF handout card
    ├── student/
    │   ├── dashboard.php         # Candidate cockpit: syllabus, tests, books, leaderboard
    │   └── profile.php           # Candidate verified dossier, subscription & logs
    └── teacher/
        └── dashboard.php         # Faculty hub: note uploads, test publishing, books
```

4. Database Architecture & Data Dictionary

All portal data is persisted in SQLite (`database/bpsc_portal.sqlite`). The schema is fully relational with foreign keys and indexes.

📊 ARCHITECTURE & SCHEMA FLOW SPECIFICATION

```
erDiagram
    USERS ||--o{ CANDIDATE_RESULTS : submits
    USERS ||--o{ PAYMENTS : initiates
    USERS ||--o{ NOTES : authors
    QUESTION_PAPERS ||--o{ CANDIDATE_RESULTS : generates
    TEACHER_APPLICATIONS ||--o| USERS : transitions_to

    USERS {
        string id PK
        string name
        string email
        string phone
        string password_hash
        string role
        string district
        string cadre_level
        string subject
        string status
        datetime created_at
    }
    QUESTION_PAPERS {
        string id PK
        string title
        string cadre
        int duration
        float total_marks
        float penalty
        string visibility
        string status
        text questions_json
        string author
        datetime published_at
    }
    QUESTION_BANK {
        int id PK
```

```
        string topic
        text question
        text options_json
        int correct_answer
        float marks
        float penalty
        text explanation
    }
    NOTES {
        int id PK
        string topic_id
        string cadre
        string subject
        string title
        text summary
        text body
        text attached_file_json
        text quiz_json
        string author
        datetime uploaded_at
    }
    CANDIDATE_RESULTS {
        int id PK
        string user_id FK
        string paper_id FK
        float score
        int correct_count
        int wrong_count
        int blank_count
        float accuracy
        datetime completed_at
    }
    PAYMENTS {
        string id PK
        string user_id FK
        string plan_name
        float amount
        string status
        datetime created_at
    }
    TEACHER_APPLICATIONS {
        string id PK
        string name
        string email
        string phone
        string district
        string cadre
        string subject
        string qualification
        string status
        datetime applied_at
    }
```

5. User Roles, Authentication & Access Control

The portal implements strict Role-Based Access Control (RBAC):

Capability / Resource	Candidate (Student)	Teacher (Faculty)	Admin (AIVERSE)
Access Public & Candidate CBT Tests	Yes	Yes (Preview)	Yes (Preview)
Download SCERT Books & PYQs	Yes	Yes	Yes
Study Topic Dossiers & Take Practice Quizzes	Yes	Yes	Yes

Capability / Resource	Candidate (Student)	Teacher (Faculty)	Admin (AIVERSE)
View District / State Leaderboards	Yes	Yes	Yes
Upload Topic Study Notes & PDF Handouts	No	Yes	Yes
Bulk Import Questions from Word (`.docx`)	No	Yes	Yes
Publish Full Mock Tests (Draft / Live)	No	Yes	Yes
Publish Books to Digital Library	No	Yes	Yes
Approve / Reject Faculty Applications	No	No	Yes
View Financial Gateway Transactions Ledger	No	No	Yes
Edit System Question Bank Database	No	No	Yes

Authentication Mechanics:

- Sign in accepts **Email, Mobile Phone Number, or User ID**.
- Passwords are verified via PHP's `password_verify()` against `PASSWORD_BCRYPT` hashes.
- For new teacher registrations, applications default to `pending_approval`. Only when approved by an administrator in the admin dashboard is the account granted login rights.

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6. Core Functional Modules & Workflows

6.1 Candidate Dashboard & Master Syllabus Engine

- **Cadre Selector**: Switches instantly between all four BPSC teacher cadres:

1. *Primary (Classes 1–5)*: 30Q Language Qualifying + 120Q General Studies & EVS.
2. *Middle (Classes 6–8)*: 30Q Language + 40Q GS Core + 80Q Concerned Subject.
3. *Secondary (Classes 9–10)*: 30Q Language + 40Q GS Core + 80Q Discipline Domain.
4. *Higher Secondary (Classes 11–12)*: 30Q Language + 40Q GS Core + 80Q Specialized Discipline.

- **6-Axis Knowledge Radar**: SVG radar chart calculating live syllabus mastery across 6 domains: Language, Elementary Math, General Science, Indian National Movement (INM) & Bihar GK, Geography, and Part III Discipline.
- **Topic Cards**: Displays completion status (`Unread` , `In Progress` , `Mastered`), high-yield indicators (`HY`), attached PDF handout badges, and quick links to read notes or take topic tests.

6.2 Proctored CBT Examination Engine & Anti-Cheat Guard

- **Pre-Exam Instructions**: Shows test duration, question count, marking rules (+1.00 / -0.33), language qualification threshold (30%), and candidate undertaking agreement.
- **Active Exam Interface**:
- Split-screen layout (desktop) and mobile sticky live HUD.
- Live countdown timer running on accurate 1-second ticks.
- Interactive radio option cards with full touch targets.
- Question navigation palette with color indicators: green (answered) and neutral (unattempted).
- **Anti-Cheat Monitor**: Binds to browser `visibilitychange` and window `blur` events. If a candidate leaves the exam tab or switches applications, the system automatically submits the exam immediately to uphold exam integrity.
- **Post-Exam Scorecard**: Calculates total merit score, correct count, wrong count, blank count, penalty deductions, and presents an interactive question-by-question solution review with explanations.

6.3 Topic Study Dossiers & Mastery Quizzes

- Comprehensive study guides covering historical, scientific, and mathematical frameworks.
- Direct download card for faculty-uploaded PDF documents and handouts.
- Focused 4-question retention quiz. Scoring ≥ 75% (3 of 4 correct) automatically promotes the topic to **Mastered** status in the student's study dossier.

6.4 Downloadable Books & Previous Year Papers (PYQs)

- Direct `.pdf` and `.doc` download actions for:
- Official SCERT Bihar and NCERT school textbooks (History, Science, Math, Geography).
- Official past examination question papers from BPSC TRE 1.0, TRE 2.0, and TRE 3.0 with official answer keys.

6.5 Live Bihar Merit Leaderboard System

- Filters candidate standings by **District (38 Bihar Districts)** or **State-Wide (All Bihar)**.
- Real-time search filter by candidate name or roll number.
- Podium showcase for Top 3 Toppers (Gold, Silver, Bronze badges) and individual student standing highlight card.

6.6 Teacher / Faculty Workspace

- Upload study notes and attach PDF handouts up to 50MB.
- Upload official Previous Year Papers with shift date and subject stream.
- Ingest Word `.docx` documents containing multiple choice questions with automated Mammoth.js parsing.
- Select questions from the system question bank to publish new mock tests.

6.7 Admin (AIVERSE) Governance & Approvals

- Review pending faculty onboarding applications with candidate experience, qualifications, and contact details.
- Approve or reject faculty with one-click actions.
- Review gateway transactions ledger (Razorpay / UPI / NetBanking simulation).
- Direct access to manage the master question bank database.

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7. API Endpoints & Server Handlers

Endpoint	Method	Payload / Parameters	Purpose
`login-handler.php`	`POST`	`username`, `password`, `role`	Authenticates user and initializes session
`register-handler.php`	`POST`	`name`, `email`, `phone`, `password`, `role`, `district`, `cadre`	Registers new candidate or submits teacher application
`logout.php`	`GET`	None	Terminates session and redirects to login
`tests-handler.php`	`POST`	`action`: `create_paper`, `toggle_status`, `submit_result`	Handles CBT creation, status toggles, and exam result submission
`notes-handler.php`	`POST`	`action`: `save_notes`, `topic_id`, `title`, `body`, file upload	Stores topic study notes and attached PDF files
`leaderboard-handler.php`	`GET`	`scope` (`district` / `state`), `district` (name)	Returns ranked candidates and current user standing
`admin-teacher-handler.php`	`POST`	`action`: `approve` / `reject`, `teacher_id`	Approves or rejects pending faculty applications

8. Mobile Responsiveness & Ergonomic UI Architecture

The portal is designed mobile-first and tested across viewports from 320px up to 1440px:

📊 **ARCHITECTURE & SCHEMA FLOW SPECIFICATION**

```
graph LR
    subgraph Mobile Viewports [< 640px]
        NavMobile["Compact Header + Mobile Hamburger"]
        BottomDock["Fixed Bottom Dock (Home | CBT | Books | PYQs | Profile)"]
        CBTHud["Sticky Exam HUD (Timer + Progress + Palette Toggle)"]
        CardsMobile["Responsive 2-Col HUD & Full-Width Tables (min-w-650px)"]
    end
    subgraph Desktop Viewports [>= 640px]
        NavDesktop["Full Horizontal Navigation Bar"]
        SidebarDesktop["Sticky 4-Col Palette & Scope Selectors"]
        GridsDesktop["4-Col Metrics & Split-Screen CBT Cockpit"]
    end
```

- **Touch Targets**: All interactive elements (Cadre tabs, radio buttons, navigation links) maintain a minimum touch target height of 42px–48px.
- **Scroll Optimization**: Tables are wrapped in `.overflow-x-auto` with `whitespace-nowrap` and `min-w-[620px]` – `min-w-[650px]` so data columns never squish or wrap into vertical stacks.
- **Form Usability**: Input font size is fixed to a minimum of 16px on mobile viewports to prevent unwanted browser auto-zooming on iOS Safari and Android Chrome.

9. Installation, Environment Setup & Deployment

System Requirements:

- PHP 8.1 or higher (with `pdo_sqlite` and `fileinfo` extensions enabled).
- SQLite3.
- Any modern web server (Apache, Nginx, or PHP's built-in development server).

Quick Start (Local Development):

```
# 1. Clone or navigate to the project directory
cd bpsc-tre-portal

# 2. Verify PHP installation and SQLite extension
php -v
php -m | grep sqlite

# 3. Start the local server
php -S 127.0.0.1:8000
```

Open http://localhost:8000 in your web browser.

Production Deployment (Apache / Nginx):

1. Copy the codebase to your web root (e.g. `/var/www/html/bpsc-tre-portal`).
2. Ensure write permissions on `database/` directory and `database/bpsc_portal.sqlite` (`chmod 775` or `chown www-data:www-data`).

3. For Apache: Enable `mod_rewrite` (`a2enmod rewrite`) so `.htaccess` rules apply.
4. For Nginx: Configure `fastcgi_pass` to your PHP-FPM socket and point `root` to the project directory.

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10. Default Credentials & Test Accounts

The database comes pre-seeded with ready-to-test accounts for every role:

Persona	Username / Email	Password	Role	Permissions
Candidate	`student@bpsc.local`	Any password	Student	Full candidate dashboard, syllabus, CBT engine, books & PYQ downloads
Faculty	`teacher@bpsc.local`	Any password	Teacher	Note uploader, PYQ publisher, Word importer, test curator
Administrator	`admin@bpsc.local`	Any password	Admin	Faculty approvals, payments ledger, question bank master

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Documentation compiled and verified for Infinity Learning BPSC TRE 4.0 Ecosystem.